



CHAPTER 4 Bipolar Junction Transistors

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4.1 Bipolar Junction Transistors Structure

4.2 Basic BJT Operation

4.3 BJT Characteristics and Parameters

4.1 Bipolar Junction Transistors (BJT) Structure

- The BJT is constructed with three doped semiconductor regions separated by two pn junctions, as shown in the epitaxial planar structure in Figure 4–1(a). The three regions are called **emitter**, **base**, and **collector**.
- Physical representations of the two types of BJTs are shown in Figure 4–1(b) and (c). One type consists of two n regions separated by a p region (**npn**), and the other type consists of two p regions separated by an n region (**pnp**).

The term bipolar refers to the use of both holes and electrons as current carriers in the transistor structure.

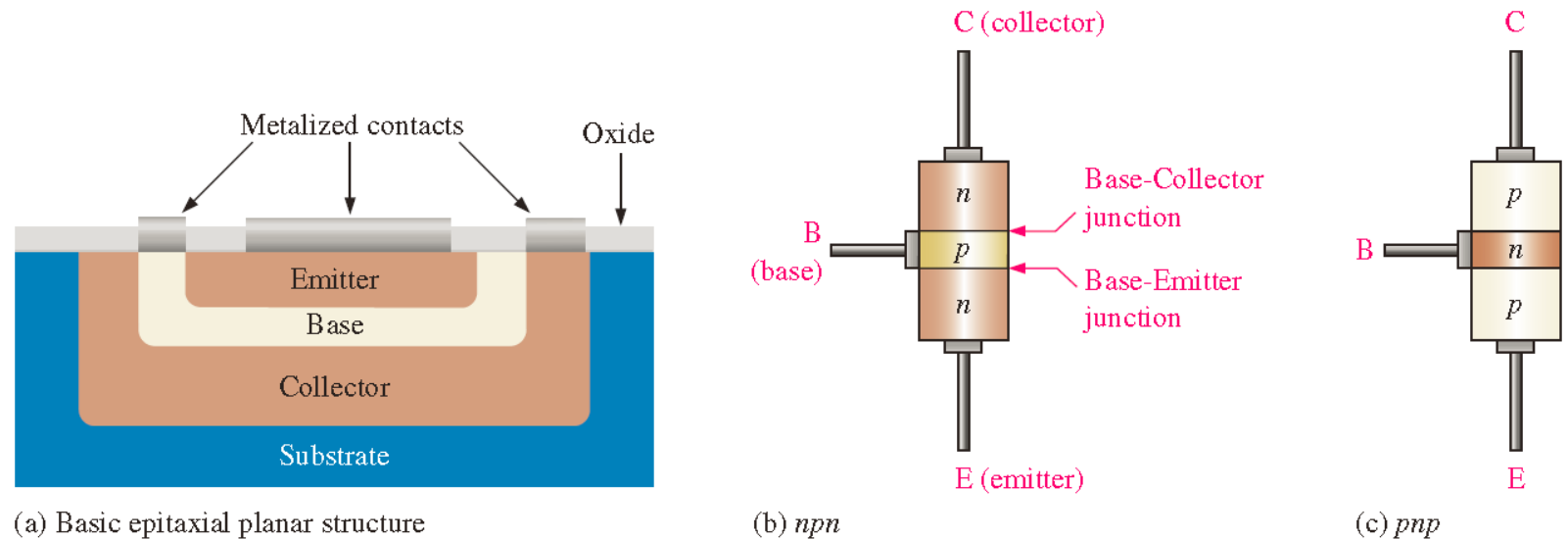
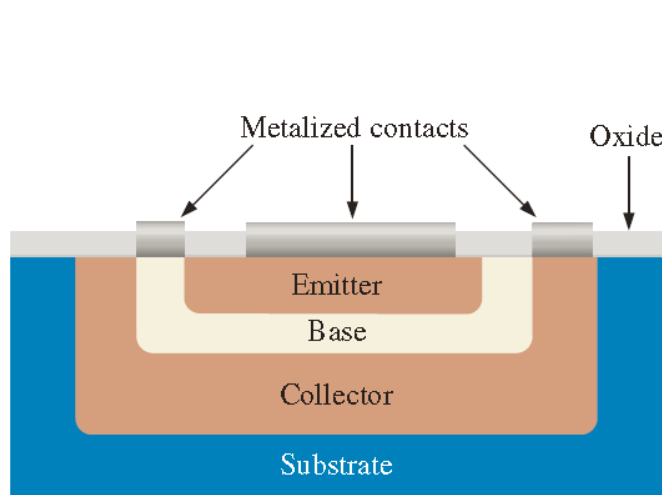


FIGURE 4-1

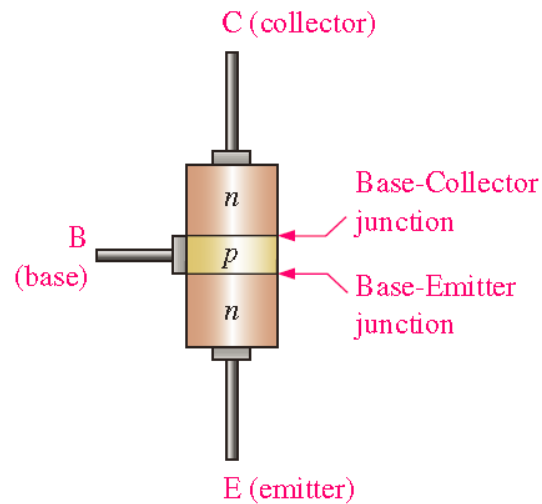
BJT construction. The substrate is a physical supporting material for the transistor.

Bipolar Junction Transistors (BJT) Structure

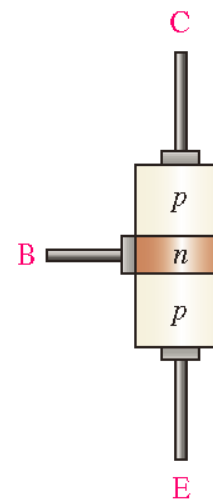
- The **pn** junction joining the base region and the emitter region is called the **base-emitter junction**. The pn junction joining the base region and the collector region is called the **base-collector junction**, as indicated in Figure 4–1(b). A lead connects to each of the three regions, as shown. These leads are labeled **E, B, and C** for **emitter, base, and collector**, respectively.
- The **base region is lightly doped** and very thin compared to the **heavily doped emitter** and the **moderately doped collector** regions.



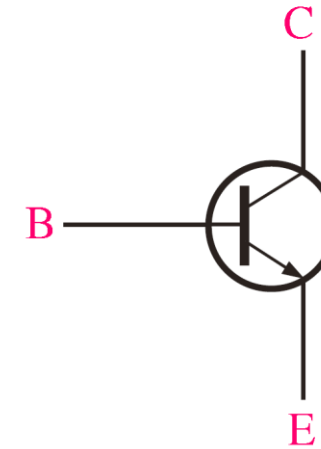
(a) Basic epitaxial planar structure



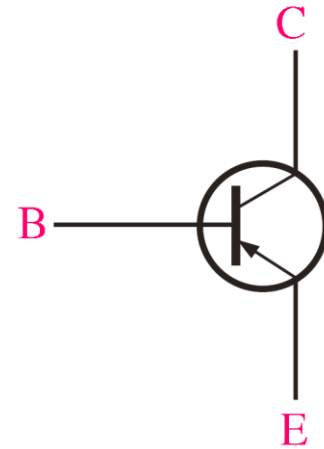
(b) npn



(c) pnp



(a) npn



(b) pnp

FIGURE 4-1

BJT construction. The substrate is a physical supporting material for the transistor.

FIGURE 4-2 Standard BJT (bipolar junction transistor) symbols

4.2 Basic BJT Operation

- Biasing:** Figure 4–3 shows a bias arrangement for both npn and pnp BJTs for operation as an amplifier. Notice that in both cases the base-emitter (BE) junction is forward-biased and the base-collector (BC) junction is reverse-biased.

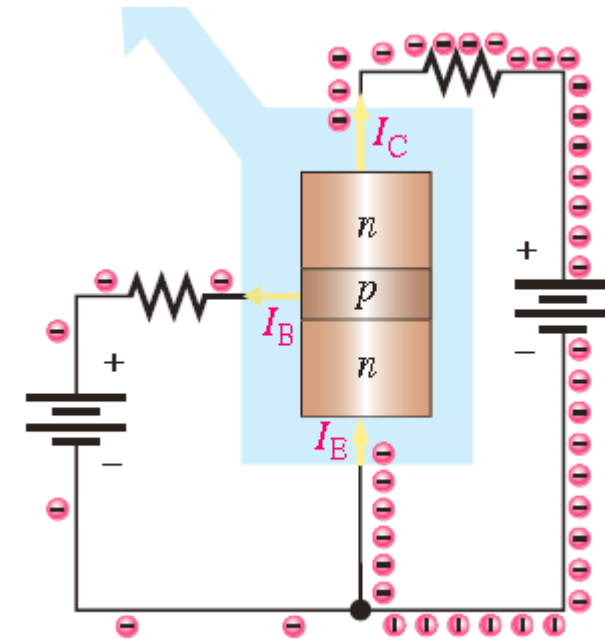
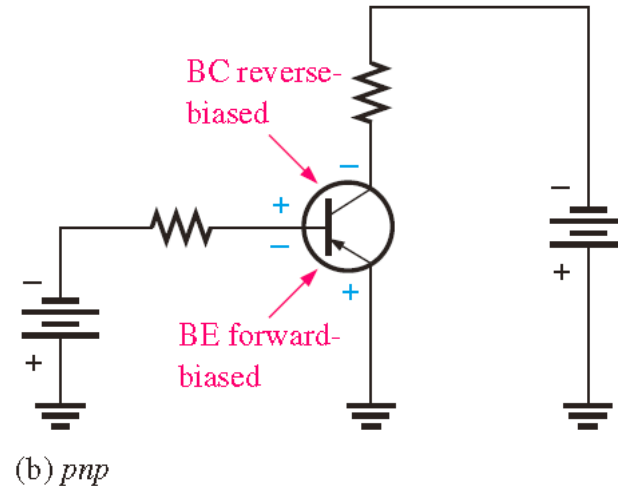
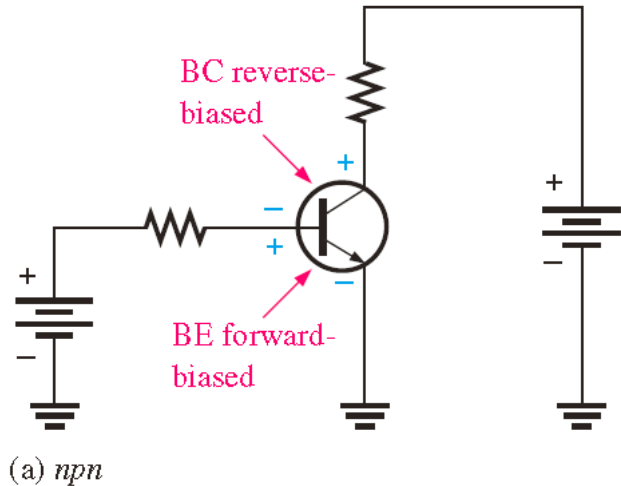


FIGURE 4-3 Forward-reverse bias of a BJT.

Basic BJT Operation

- Free electrons easily diffuse through the forward biased BE junction into the lightly doped and very thin p-type base region. A small percentage of the total number of free electrons injected into the base region recombine with holes.

The rest free electrons move toward the reverse-biased BC junction, they are swept across into the collector region by the attraction of the positive collector supply voltage.

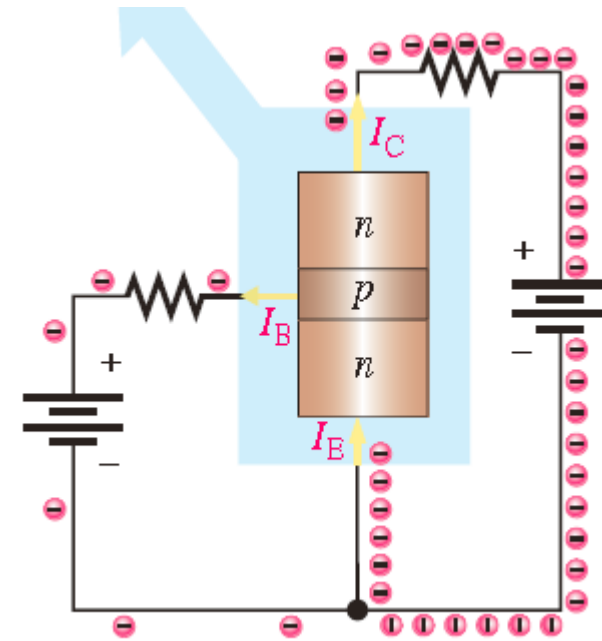


FIGURE 4-4 BJT operation showing electro flow.

Transistor Currents

- The **directions of the currents** in an npn transistor and its schematic symbol are as shown in Figure 4–5(a);
- These diagrams show that the emitter current (I_E) is the sum of the collector current (I_C) and the base current (I_B), expressed as follows:

$$I_E = I_C + I_B$$

As mentioned before, I_B is very small compared to I_E or I_C . The capital-letter subscripts indicate dc values.

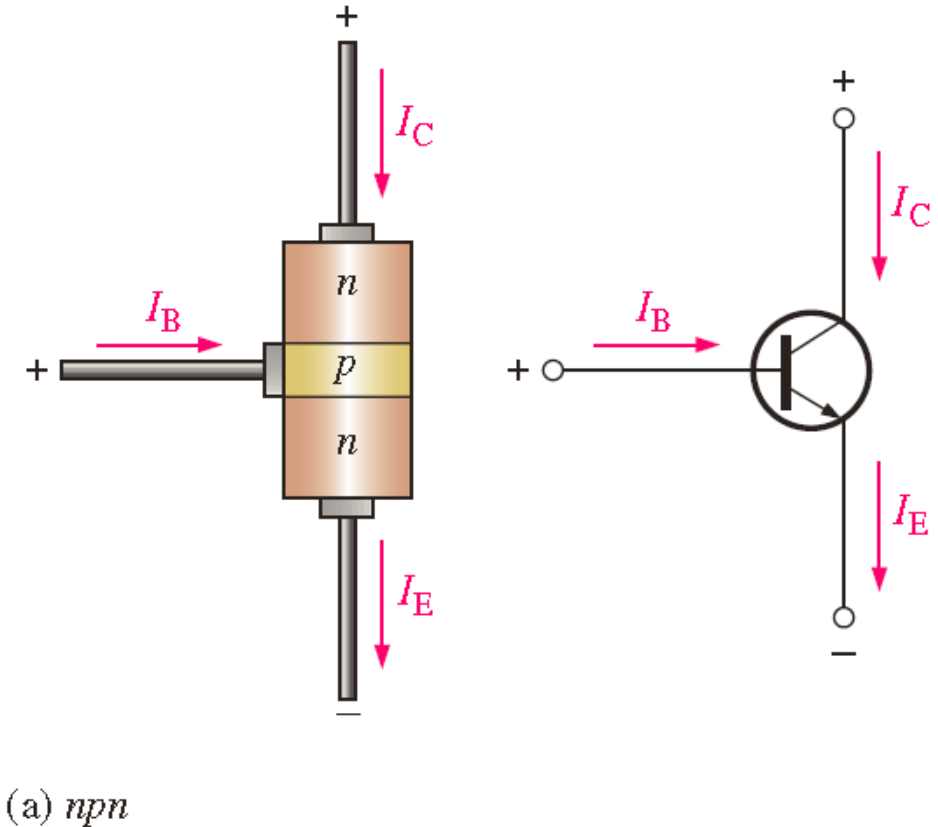


FIGURE 4-5 Transistor currents

4-3 BJT Characteristics and Parameters

I_B / mA	-0.01	0	0.02	0.04	0.06	0.08	0.1
I_C / mA	0.001	0.01	0.7	1.5	2.3	3.1	3.95
I_E / mA	0	0.01	0.72	1.54	2.36	3.18	4.05

$$I_E = I_C + I_B \quad I_C \gg I_B$$

- BJT has **current amplifier** function. Current amplify gain is almost constant within acceptable varying range.
- The dc current **gain** of a transistor is the ratio of the dc collector current (I_C) to the dc base current (I_B) and is designated dc beta (β_{DC}).

$$\beta_{DC} = \frac{I_C}{I_B}$$

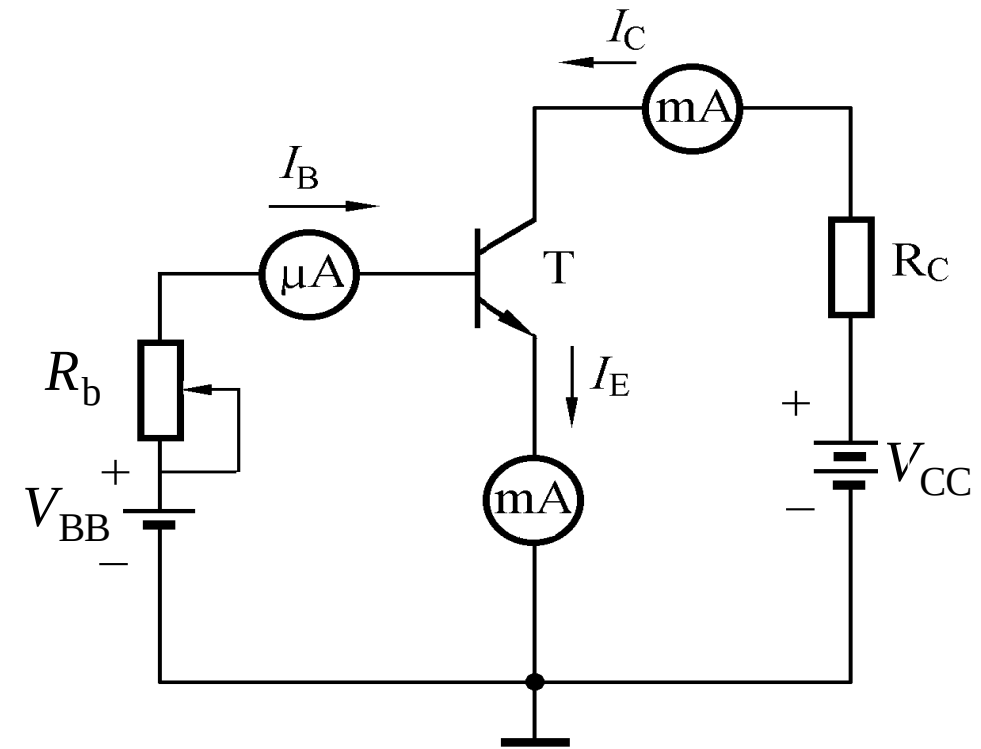


FIGURE 4-6 Testing circuit of Forward-reverse biased BJT.

Through varying the resistor, I_B , I_C and I_E can be varied correspondingly.

Example

EXAMPLE 4–1

Determine the dc current gain β_{DC} and the emitter current I_E for a transistor where $I_B = 50 \mu\text{A}$ and $I_C = 3.65 \text{ mA}$.

Solution

$$\beta_{DC} = \frac{I_C}{I_B} = \frac{3.65 \text{ mA}}{50 \mu\text{A}} = 73$$

$$I_E = I_C + I_B = 3.65 \text{ mA} + 50 \mu\text{A} = 3.70 \text{ mA}$$

*Related Problem**

A certain transistor has a β_{DC} of 200. When the base current is $50 \mu\text{A}$, determine the collector current.

Transistor DC Model

- You can view the unsaturated BJT as a device with a **current input and a dependent current source** in the output circuit, as shown in Figure 4–7 for an *npn transistor*.
- The input circuit is a **forward-biased diode** through which there is base current. The output circuit is a **dependent current source** (diamond-shaped element) with a value that is dependent on the base current, I_B , and equal to $\beta_{DC}I_B$.

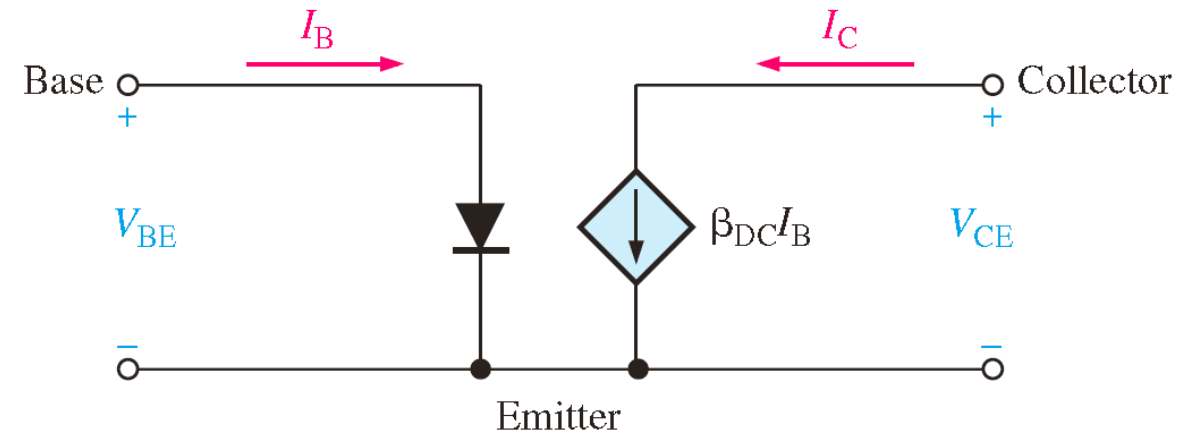


FIGURE 4-7

Ideal dc model of an *npn* transistor

BJT Circuit Analysis

I_B : dc base current

I_E : dc emitter current

I_C : dc collector current

V_{BE} : dc voltage at base with respect to emitter

V_{CB} : dc voltage at collector with respect to base

V_{CE} : dc voltage at collector with respect to emitter

The base-bias voltage source, V_{BB} , forward-biases the base-emitter junction, and the collector-bias voltage source, V_{CC} , reverse-biases the base-collector junction. When the base-emitter junction is forward-biased, it is like a forward-biased diode and has a nominal forward voltage drop of $V_{BE} \cong 0.7 \text{ V}$

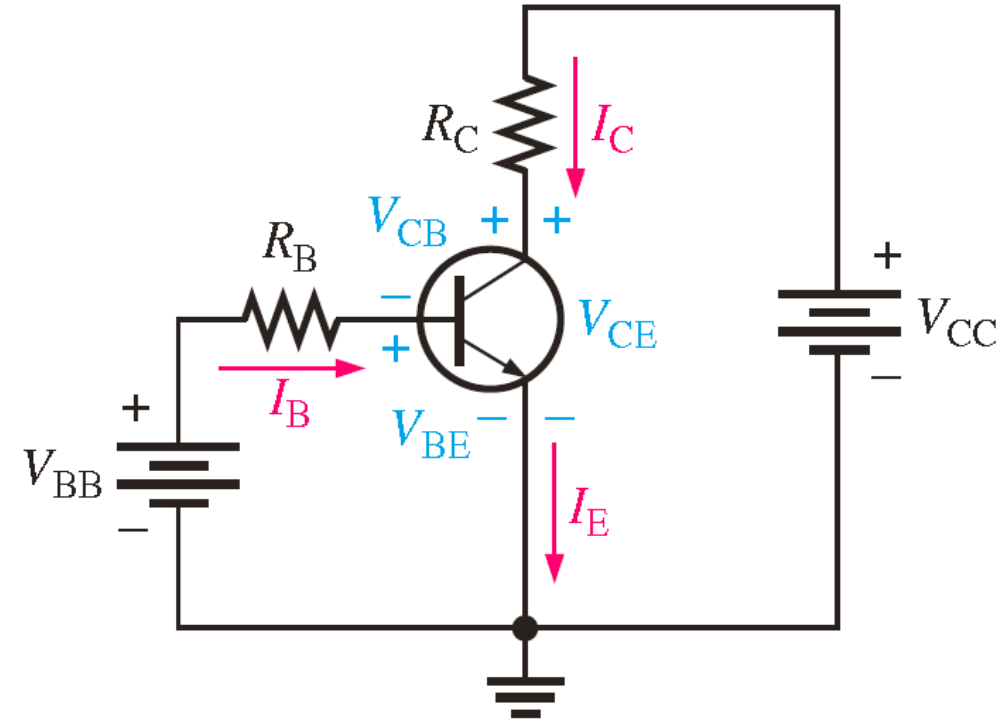


FIGURE 4-8

Transistor currents and voltages.

BJT Circuit Analysis

Since the emitter is at ground (0 V), by Kirchhoff's voltage law, the voltage across R_B is

$$V_{R_B} = V_{BB} - V_{BE}$$

Also, by Ohm's law,

$$V_{R_B} = I_B R_B$$

Substituting for V_{R_B} yields

$$I_B R_B = V_{BB} - V_{BE}$$

Solving for I_B ,

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

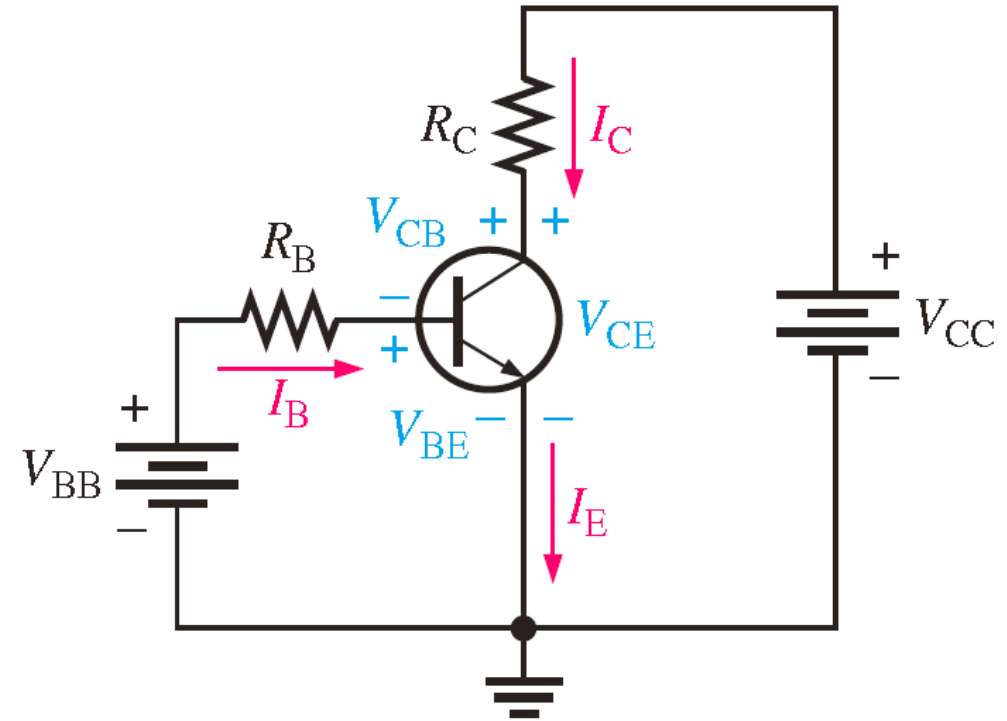


FIGURE 4-8

Transistor currents and voltages.

BJT Circuit Analysis

The voltage at the collector with respect to the grounded emitter is

$$V_{CE} = V_{CC} - V_{R_C}$$

Since the drop across R_C is

$$V_{R_C} = I_C R_C$$

the voltage at the collector with respect to the emitter can be written

$$V_{CE} = V_{CC} - I_C R_C$$

where $I_C = \beta_{DC} I_B$.

The voltage across the reverse-biased collector-base junction is

$$V_{CB} = V_{CE} - V_{BE}$$

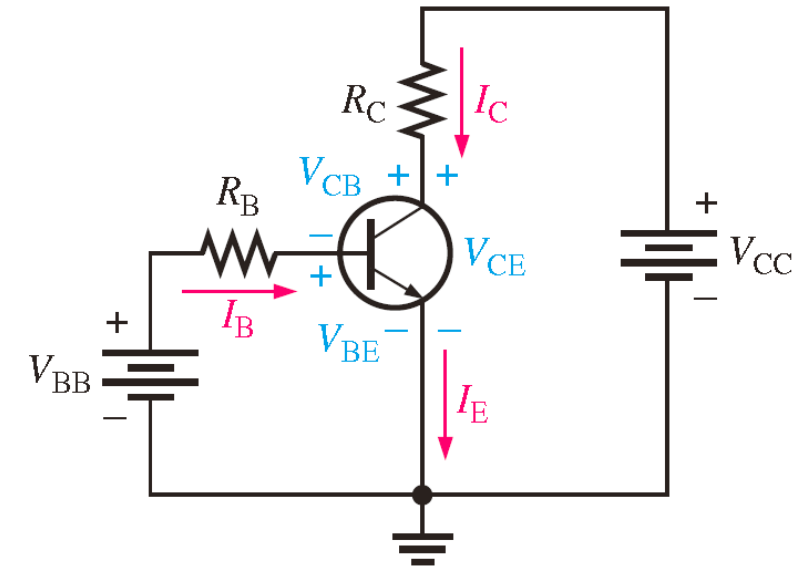


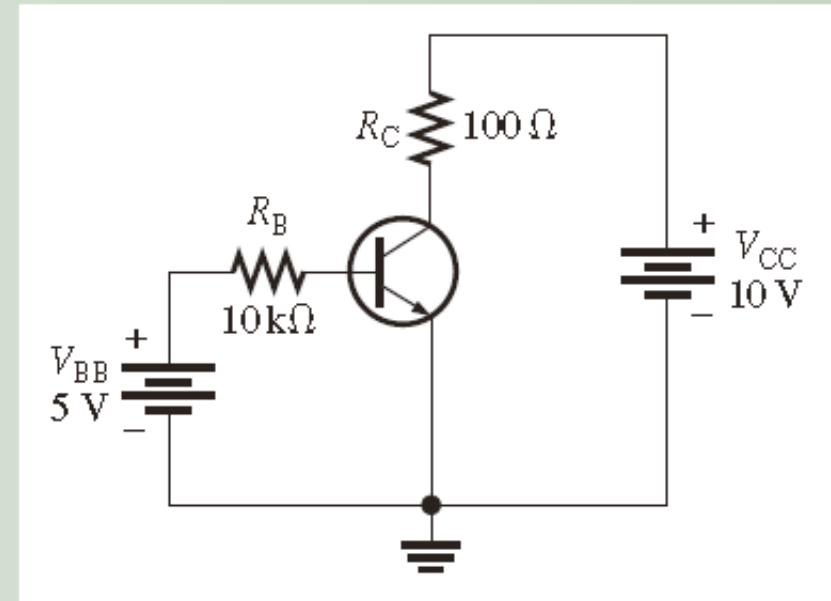
FIGURE 4-8
Transistor currents and voltages.

Example

EXAMPLE 4–2

Determine I_B , I_C , I_E , V_{BE} , V_{CE} , and V_{CB} in the circuit of Figure 4–9. The transistor has a $\beta_{DC} = 150$.

► FIGURE 4–9



Example

Solution From Equation 4–3, $V_{BE} \cong 0.7 \text{ V}$. Calculate the base, collector, and emitter currents as follows:

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{5 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega} = 430 \mu\text{A}$$

$$I_C = \beta_{DC} I_B = (150)(430 \mu\text{A}) = 64.5 \text{ mA}$$

$$I_E = I_C + I_B = 64.5 \text{ mA} + 430 \mu\text{A} = 64.9 \text{ mA}$$

Solve for V_{CE} and V_{CB} .

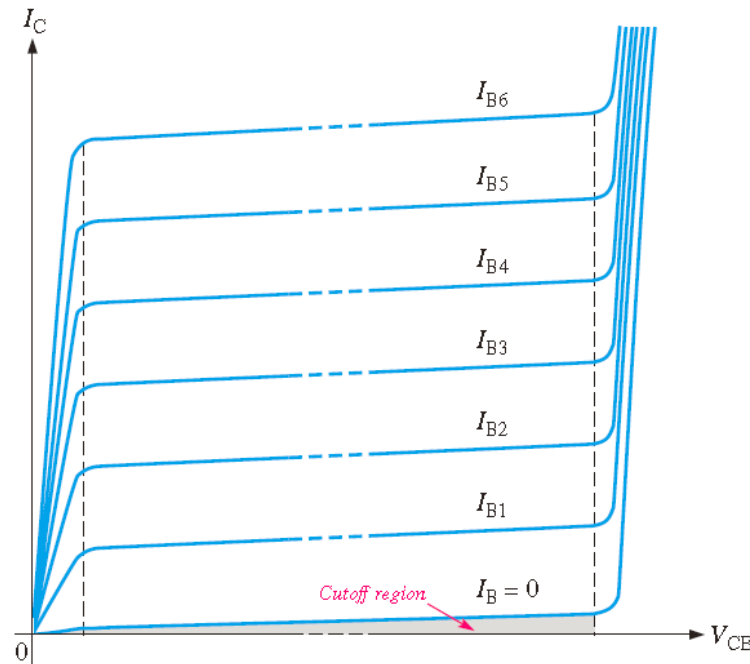
$$V_{CE} = V_{CC} - I_C R_C = 10 \text{ V} - (64.5 \text{ mA})(100 \Omega) = 10 \text{ V} - 6.45 \text{ V} = 3.55 \text{ V}$$

$$V_{CB} = V_{CE} - V_{BE} = 3.55 \text{ V} - 0.7 \text{ V} = 2.85 \text{ V}$$

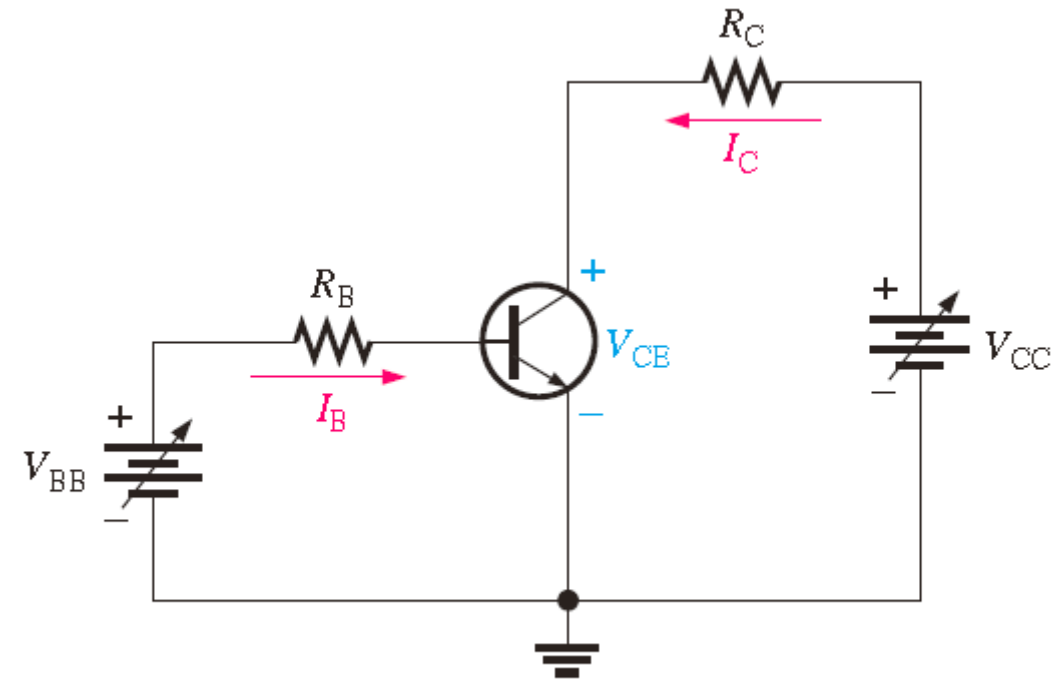
Since the collector is at a higher voltage than the base, the collector-base junction is reverse-biased.

Collector Characteristic Curves

- Using a circuit like that shown in Figure 4–10(a), a set of collector characteristic curves can be generated that show how the collector current, I_C , varies with the collector-to-emitter voltage, V_{CE} , for specified values of base current.



(c) Family of I_C versus V_{CE} curves for several values of I_B
($I_{B1} < I_{B2} < I_{B3}$, etc.)



(a) Circuit

FIGURE 4-10

Collector characteristic curves.

Collector Characteristic Curves

- V_{CE} is near 0 V. When both junctions are forward-biased, the transistor is in the **saturation** region of its operation. Saturation is the state of a BJT in which the collector current has reached a maximum and is independent of the base current.
- As V_{CC} is increased, V_{CE} increases as the collector current increases. This is indicated by the portion of the characteristic curve between points A and B in Figure 4–10(b).
- I_C increases as V_{CC} is increased because V_{CE} remains less than 0.7 V due to the forward-biased base-collector junction.

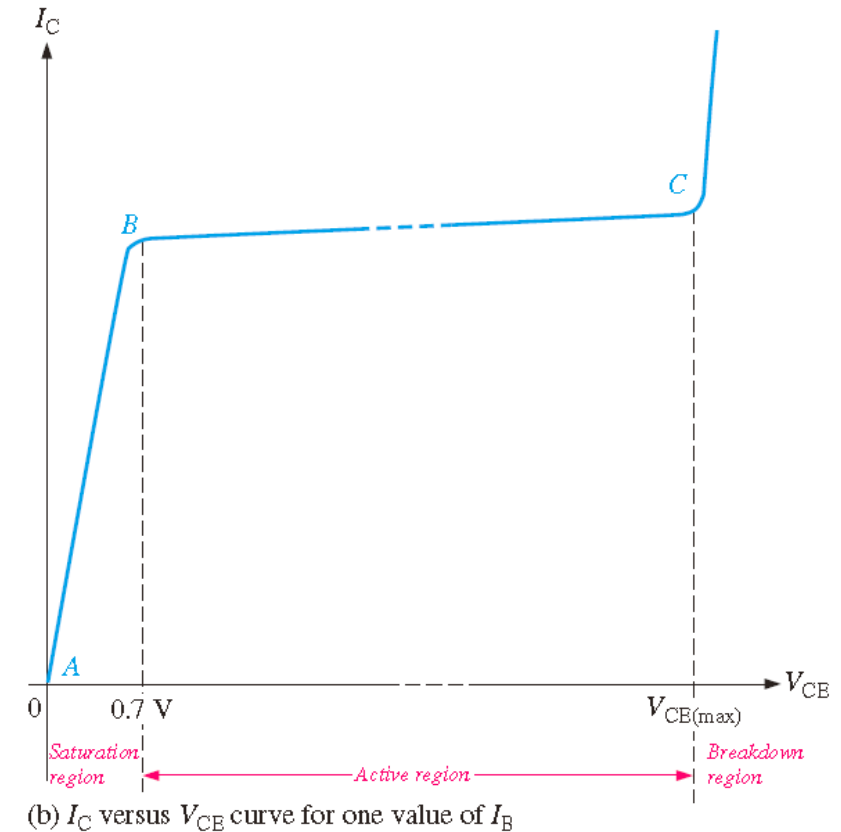


FIGURE 4-10

Collector characteristic curves.

Collector Characteristic Curves

- Ideally, when V_{CE} exceeds 0.7 V, the base-collector junction becomes reverse-biased and the transistor goes into the active, or linear, region of its operation.
- Once the base collector junction is reverse-biased, I_C levels off and remains essentially constant for a given value of I_B as V_{CE} continues to increase.
- This is shown by the portion of the characteristic curve between points B and C in Figure 4–10(b). For this portion of the characteristic curve, the value of I_C is determined only by the relationship expressed as
 - $I_C = \beta_{DC} I_B$.

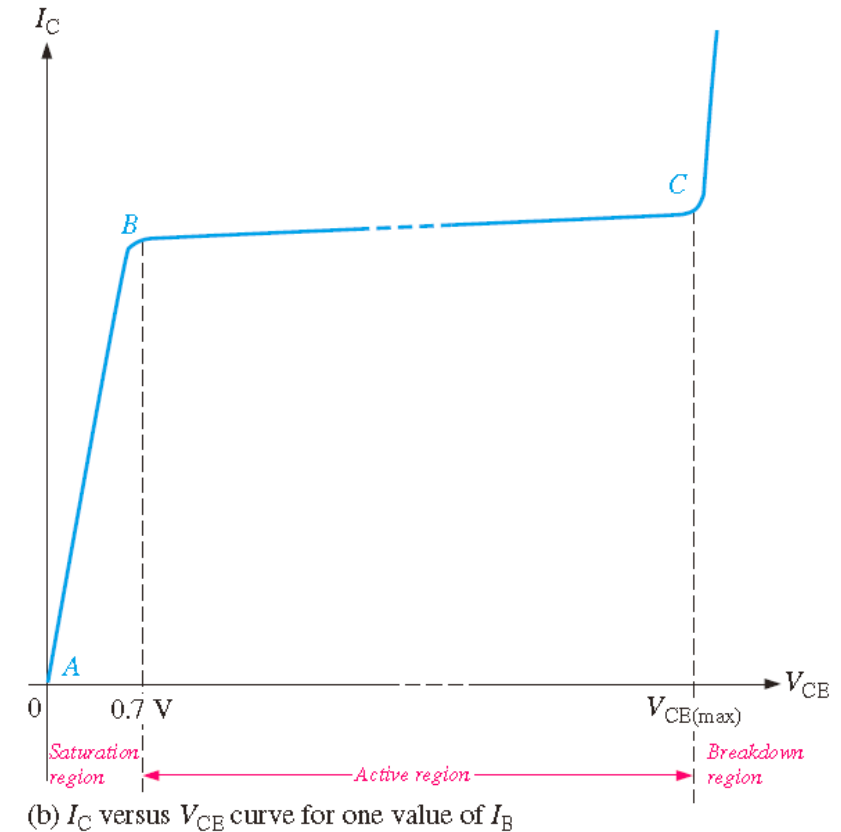


FIGURE 4-10

Collector characteristic curves.

Collector Characteristic Curves

- When V_{CE} reaches a sufficiently high voltage, the reverse-biased base-collector junction goes into breakdown; and the collector current increases rapidly as indicated by the part of the curve to the right of point C in Figure 4–10(b). A transistor should never be operated in this **breakdown region**.
- When $I_B = 0$, the transistor is in the **cutoff region** although there is a very small collector leakage current as indicated. Cutoff is the nonconducting state of a transistor.

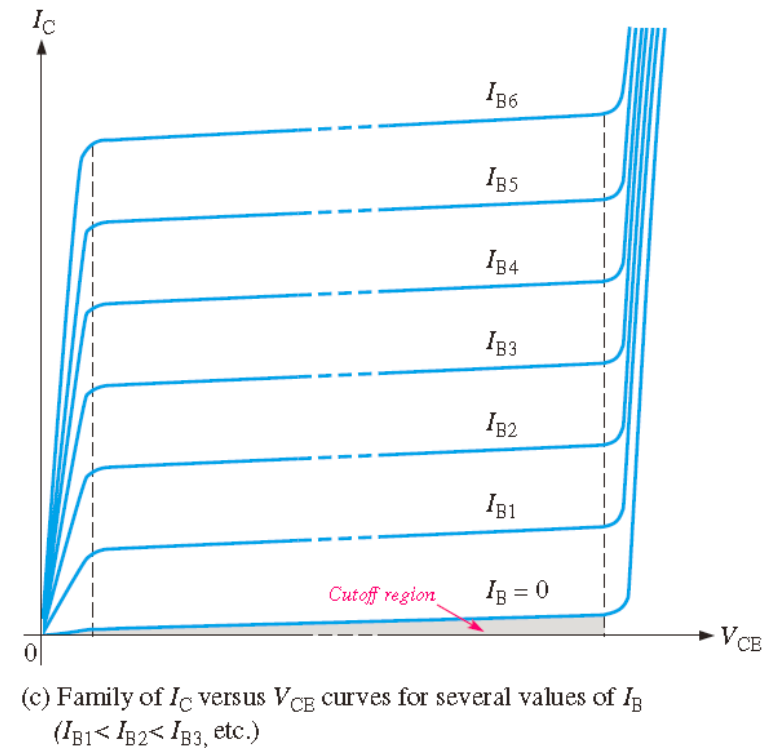


FIGURE 4-10

Collector characteristic curves.

Cutoff

- As previously mentioned, when $I_B = 0$, the transistor is in the cutoff region of its operation. This is shown in Figure 4–13 with the base lead open, resulting in a base current of zero.
- Under this condition, there is a very small amount of collector leakage current, I_{CEO} , due mainly to thermally produced carriers. Because I_{CEO} is extremely small, it will usually be neglected in circuit analysis so that $V_{CE} = V_{CC}$. In cutoff, neither the base-emitter nor the base-collector junctions are forward-biased.

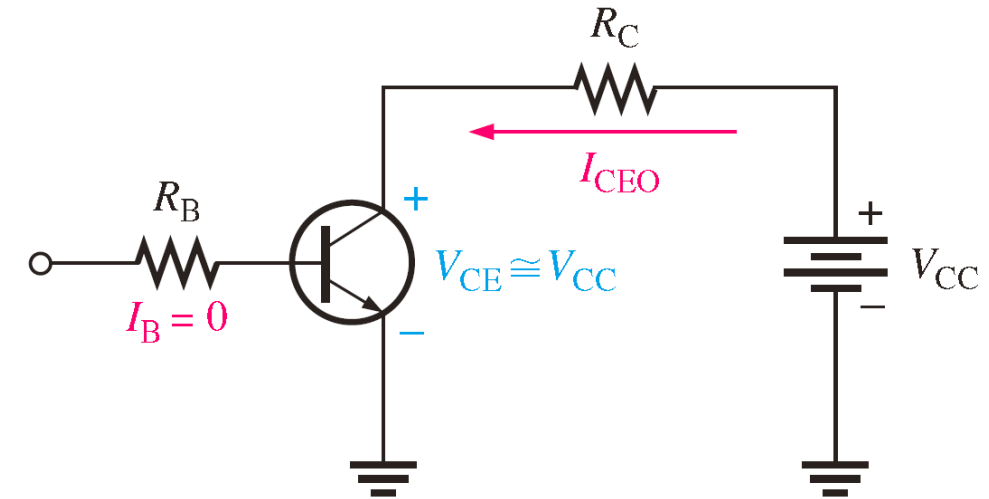


FIGURE 4-11

Cutoff: Collector leakage current (I_{CEO}) is extremely small and is usually neglected. Base-emitter and basecollector junctions are reverse-biased.

Saturation

- When the base-emitter junction becomes forward-biased and the base current is increased, the collector current also increases ($I_C = \beta_{DC} I_B$) and V_{CE} decreases as a result of more drop across the collector resistor ($V_{CE} = V_{CC} - I_C R_C$). This is illustrated in Figure 4–12.
- When V_{CE} reaches its saturation value, $V_{CE(sat)}$, the base-collector junction becomes forward-biased and I_C can increase no further even with a continued increase in I_B .
- At the point of saturation, the relation $I_C = \beta_{DC} I_B$ is no longer valid. $V_{CE(sat)}$ for a transistor occurs somewhere below the knee of the collector curves, and it is usually only a few tenths of a volt.

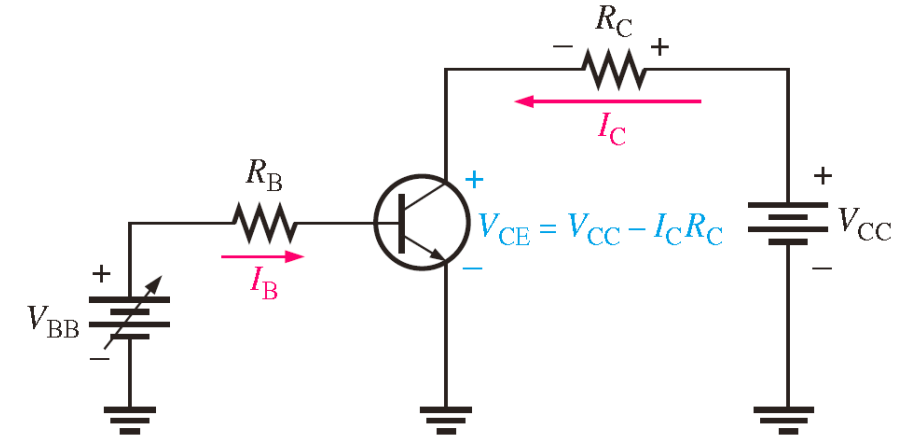


FIGURE 4-12

Saturation: As I_B increases due to increasing V_{BB} , I_C also increases and V_{CE} decreases due to the increased voltage drop across R_C . When the transistor reaches saturation, I_C can increase no further regardless of further increase in I_B . Base-emitter and base-collector junctions are forward-biased.